HSR–Stud Document Classes

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May 25, 2023

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1 Purpose of these document classes

The following document classes were made by and for the HSR-Stud organization. These classes are intended to be used to create beautiful summaries and cheat sheets. Available classes are:

`hsrzf` For summaries (Zusammenfassung)

`hsrbericht` For reports (Bericht)

2 Class Options

`sans` Use Sans-Serif fonts (`cmbright` and `hfbright`).

`slab` Use Slab-Serif fonts (`bitter` and `eulervm`).

`margin=<size>` Sets up margins to reasonable size.

`*` Any other given option is passed to the underlying class (article).

Size	H	V	Size	H	V
huge	45	40	small	20	20
large	35	35	tiny	10	20
big	30	30	minimal	5	20
normal	25	30			

Table 1: Sizes for the `margin` option. All sizes are in millimeters

3 Extra Metadata

Thanks to the `titling` package, normal metadata informations are available throug the following commands: `\theauthor`, `\thedate`, `\thetitle`. Additionally, more metadata variables have been created for this class, to conserve informations related to the university course.

`\course` The degree programme of the discussed subject. Example: `Elektrotechnik`

`\module` Short name of the module according to the class schedule. Example: `ComEng1`

`\semester` Semester during which the summarized subject was taught. Example: `FS20`

`\institute` Internal institute in which the module was taught. This field is optional.

`\authoremail` Email of the author. This field is optional.

`\maintainer` Name of the person maintaining the work. This field is optional.

`\contributors` List of names of those who have contributed to the document. This field is optional.

Like `\date`, `\title`, etc. these fields have too an accessor macro `\the<metadata>`. For example: `\thecourse`, `\themodule`, and so on.

4 Zusammenfassung `hsrzf` class

This class is intended to be used to write short summaries.

4.1 Class options

`header` Use the HSR–Stud standard header. The option `noheader` disables the header.

5 Bericht `hsrbericht` class

This class tries to conform to the indications given by the communication department of the HSR. The specifications are found in the “Werkzeugkastens Technische Berichte”.

5.1 Metadata

In addition to the metadata described in the previous section. This class requires:

`\subtitle` Pretty self explanatory.

`\category` Category assigned to the project. This field is optional.

`\project` Name or reference to your project.

`\topic` You chosen topic in the category.

\instructor Name of the instructor / academic that is following or has assigned your work.

\startdate Date when you started the project.

\finishdate Date when you finished the project (can be set to **\today** to keep automatically updated.

\address Address or place where your work is produced. This field is optional.

\matriculationnr Your matriculation number.

\logo Code to insert your logo.

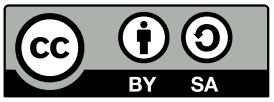
\clientlogo Code to insert the logo of the client / research department / ….

6 Minimal working examples

```
%  \documentclass[margin=normal]{hsrzf}
%
%  % Packages
%  %% the HSRStud package
%  \usepackage{hsrstud}
%
%  % Metadata
%  \course{Elektrotechnik}
%  \module{ModAbk}
%  \semester{Herbstsemester 2020}
%
%  \authoremail{vname@hsr.ch}
%  \author{\textsl{Vorname Name} -- \texttt{\theauthoremail}}
%
%  \title{\texttt{\themodule} Zusammenfassung}
%  \date{\thesemester}
%
%  % document
%  \begin{document}
%  \maketitle
%  \tableofcontents
%  \section{Introduction}
%  \[
%    i\hbar \partial_t \left| \psi(t) \right\rangle
%      = \hat{H} \left| \psi(t) \right\rangle
%  \]
%
%  \end{document}
```

7 License

This work is licensed under a Creative Commons "Attribution-ShareAlike 4.0 International" license.



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Numbers written in italic refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

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8 Implementation

```
%^^A%%%
```

```
\RequirePackage{etoolbox}

%% Default font
\RequirePackage{lmodern}

%% Class options
\SetupKeyvalOptions{
  family=hsr,
  prefix=hsr@
}

%% Use header and footer
\DeclareBoolOption[false]{header}
\DeclareComplementaryOption{noheader}{header}

%% Use a sans-serif slab-serif or concrete font
\DeclareBoolOption[false]{sans}
\DeclareBoolOption[false]{slab}
\DeclareBoolOption[false]{concrete}

%% Configures the geometry package
%%
%% Possible values:
%%  - huge    (45mm, 40mm)
%%  - large   (35mm, 35mm)
%%  - big     (30mm, 30mm)
%%  - normal  (25mm, 30mm)
%%  - small   (20mm, 20mm)
%%  - tiny    (10mm, 20mm)
%%  - minimal ( 5mm, 20mm)
\DeclareStringOption{margin}[huge]

%% Configures the polyglossia package
% \DeclareStringOption[]{lang}[german]


%% Pass other options to the class
\DeclareDefaultOption{%
  \ifx\CurrentOptionValue\relax
    \PackageWarningNoLine{\@currname}{%
      Unknown option ``\CurrentOption''\MessageBreak
      is passed to class ``article''%
    }%
    % Pass the option to package color.
    % Again it is better to expand \CurrentOption.
    \expandafter\PassOptionsToClass
      \expandafter{\CurrentOption}{article}
  \else
    % Package color does not take options with values.
    % We provide the standard LaTeX error.
    \@unknownoptionerror
  \fi
}

\ProcessKeyvalOptions{hsr}


%% Create new metadata fields
```

```
%% Required fields
\newcommand\course[1]{\gdef\hsr@course{#1}}
\newcommand\hsr@course{\ClassError{hsrzf}{no \noexpand\course given}{}}
\newcommand\thecourse{\hsr@course}

\newcommand\module[1]{\gdef\hsr@module{#1}}
\newcommand\hsr@module{\ClassError{hsrzf}{no \noexpand\module given}{}}
\newcommand\themodule{\hsr@module}

\newcommand\semester[1]{\gdef\hsr@semester{#1}}
\newcommand\hsr@semester{\ClassError{hsrzf}{no \noexpand\semester given}{}}
\newcommand\thesemester{\hsr@semester}

%% Optional fields
\newcommand\institute[1]{\gdef\hsr@institute{#1}}
\newcommand\hsr@institute{\ClassWarning{hsrzf}{no \noexpand\institute given}{}}
\newcommand\theinstitute{\hsr@institute}

\newcommand\authoremail[1]{\gdef\hsr@authoremail{#1}}
\newcommand\hsr@authoremail{\ClassWarning{hsrzf}{no \noexpand\authoremail given}{}}
\newcommand\theauthoremail{\hsr@authoremail}

\newcommand\maintainer[1]{\gdef\hsr@maintainer{#1}}
\newcommand\hsr@maintainer{\ClassWarning{hsrzf}{no \noexpand\maintainer given}{}}
\newcommand\themaintainer{\hsr@maintainer}

\newcommand\contributors[1]{\gdef\hsr@contributors{#1}}
\newcommand\hsr@contributors{\ClassWarning{hsrzf}{no \noexpand\contributors given}{}}
\newcommand\thecontributors{\hsr@contributors}
⟨/hsrzf | hsrbericht⟩
⟨*hsrbericht⟩
%% The report class has some extra metadata

%% Extra title informations
\newcommand\subtitle[1]{\gdef\hsr@subtitle{#1}}
\newcommand\hsr@subtitle{\ClassError{hsrbericht}{no \noexpand\subtitle given}{}}
\newcommand\thesubtitle{\hsr@subtitle}

%% Project informations
\newcommand\category[1]{\gdef\hsr@category{#1}}
\newcommand\hsr@category{\ClassWarning{hsrbericht}{no \noexpand\category given}{}}
\newcommand\thecategory{\hsr@category}

\newcommand\project[1]{\gdef\hsr@project{#1}}
\newcommand\hsr@project{\ClassError{hsrbericht}{no \noexpand\project given}{}}
\newcommand\theproject{\hsr@project}

\newcommand\topic[1]{\gdef\hsr@topic{#1}}
\newcommand\hsr@topic{\ClassError{hsrbericht}{no \noexpand\topic given}{}}
\newcommand\thetopic{\hsr@topic}

\newcommand\version[1]{\gdef\hsr@version{#1}}
\newcommand\hsr@version{\ClassError{hsrbericht}{no \noexpand\version given}{}}
\newcommand\theversion{\hsr@version}

\newcommand\instructor[1]{\gdef\hsr@instructor{#1}}
\newcommand\hsr@instructor{\ClassError{hsrbericht}{no \noexpand\instructor given}{}}
```

```
\newcommand\theinstructor{\hsr@instructor}

\newcommand\startdate[1]{\gdef\hsr@startdate{#1}}
\newcommand\hsr@startdate{\ClassError{hsrbericht}{no \noexpand\startdate given}{}}
\newcommand\thestartdate{\hsr@startdate}

\newcommand\finishdate[1]{\gdef\hsr@finishdate{#1}}
\newcommand\hsr@finishdate{\ClassError{hsrbericht}{no \noexpand\finishdate given}{}}
\newcommand\thefinishdate{\hsr@finishdate}

%% Author informations
\newcommand\address[1]{\gdef\hsr@address{#1}}
\newcommand\hsr@address{\ClassWarning{hsrbericht}{no \noexpand\address given}{}}
\newcommand\theaddress{\hsr@address}

\newcommand\matriculationnr[1]{\gdef\hsr@matriculationnr{#1}}
\newcommand\hsr@matriculationnr{\ClassError{hsrbericht}{no \noexpand\matriculationnr given}{}}
\newcommand\thematriculationnr{\hsr@matriculationnr}

%% Logos informations
\newcommand\logo[1]{\gdef\hsr@logo{#1}}
\newcommand\hsr@logo{\ClassWarning{hsrbericht}{no \noexpand\logo given}{}}
\newcommand\thelogo{\hsr@logo}

\newcommand\clientlogo[1]{\gdef\hsr@clientlogo{#1}}
\newcommand\hsr@clientlogo{\ClassWarning{hsrbericht}{no \noexpand\clientlogo given}{}}
\newcommand\theclientlogo{\hsr@clientlogo}
⟨/hsrbericht⟩
⟨∗hsrzf | hsrbericht⟩

%^^A%%%
```

```
    sans and concrete options cannot be used at the same time
  }
  \fi

  \PassOptionsToPackage{T1}{fontenc}
  \RequirePackage{fontenc}
  \RequirePackage{cmbright}
  \RequirePackage{sfmath}
\fi

%% Configure Slab-Serif font
\ifhsr@slab
  \ifhsr@sans
    \PackageError{hsrzf}{%
      slab and sans options cannot be used at the same time
    }
  \fi
  \ifhsr@concrete{hsrzf}{%
    slab and concrete options cannot be used at the same time
  }
  \fi

  \PassOptionsToPackage{T1}{fontenc}
  \RequirePackage{fontenc}
  \PassOptionsToPackage{rm,light}{roboto}
  \RequirePackage{roboto}
\fi

%% Configure (secret) concrete font
\ifhsr@concrete
  \ifhsr@sans{hsrzf}{%
    concrete and sans options cannot be used at the same time
  }
  \fi
  \ifhsr@slab
    \PackageError{hsrzf}{%
      concrete ans slab options cannot be used at the same time
    }
  \fi

  \PassOptionsToPackage{boldsans,amssymb}{concmath}
  \RequirePackage{concmath}
  \PassOptionsToPackage{T1}{fontenc}
  \RequirePackage{fontenc}
\fi

%% configure geometry
%% defaults to big
\newlength{\hsr@hmarginwidth}
\newlength{\hsr@vmarginheight}

\ifx\hsr@margin\@empty
  \setlength\hsr@hmarginwidth{30mm}
  \setlength\hsr@vmarginheight{30mm}
\else
  \expandafter\ifstrequal\expandafter{\hsr@margin}{huge}{
    \setlength\hsr@hmarginwidth{45mm}
    \setlength\hsr@vmarginheight{40mm}}{}
```

```
%%
  \expandafter\ifstrequal\expandafter{\hsr@margin}{large}{
    \setlength\hsr@hmarginwidth{35mm}
    \setlength\hsr@vmarginheight{35mm}}{}
%%
  \expandafter\ifstrequal\expandafter{\hsr@margin}{big}{
    \setlength\hsr@hmarginwidth{30mm}
    \setlength\hsr@vmarginheight{30mm}}{}
%%
  \expandafter\ifstrequal\expandafter{\hsr@margin}{normal}{
    \setlength\hsr@hmarginwidth{25mm}
    \setlength\hsr@vmarginheight{30mm}}{}
%%
  \expandafter\ifstrequal\expandafter{\hsr@margin}{small}{
    \setlength\hsr@hmarginwidth{20mm}
    \setlength\hsr@vmarginheight{20mm}}{}
%%
  \expandafter\ifstrequal\expandafter{\hsr@margin}{tiny}{
    \setlength\hsr@hmarginwidth{10mm}
    \setlength\hsr@vmarginheight{20mm}}{}
%%
  \expandafter\ifstrequal\expandafter{\hsr@margin}{minimal}{
    \setlength\hsr@hmarginwidth{ 5mm}
    \setlength\hsr@vmarginheight{20mm}}{}
%%
  % TODO: fix this
  % \ClassError{hsrzf}{Invalid margin size}
  % \setlength\hsr@hmarginwidth{30mm}
  % \setlength\hsr@vmarginheight{30mm}
\fi

⟨/hsrzf | hsrbericht⟩

%^^A%%%
```

```
\fancypagestyle{hsrzf-norulers}{%
    \fancyhf{} % clear
    \fancyhead[L]{\leftmark}
    \fancyhead[C]{\themodule}
    \fancyhead[R]{\@date}

    \fancyfoot[L]{\href{https://github.com/HSR-Stud/}{\texttt{HSR-Stud}}}
    \fancyfoot[C]{\texttt{CC BY-NC-SA}}
    \fancyfoot[R]{\thepage ~/ \pageref{TotPages}}

    \renewcommand{\headrulewidth}{0pt}
    \renewcommand{\footrulewidth}{0pt}
}

\ifhsr@header
\pagestyle{hsrzf}
\fi
⟨/hsrzf⟩

%^^A%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%^^A HSR Bericht specific
⟨*hsrbericht⟩

%% Redefine title and abstract
\renewcommand\maketitle{%
\begin{titlepage}
    \thispagestyle{hsrbericht}
    \begin{center}
    \vspace*{6cm}
    \textbf{\fontsize{48}{60}\selectfont \@title}
    \vspace{1.5cm} \\
    \textbf{\fontsize{24}{48}\selectfont \hsr@subtitle}
    \vspace{1cm} \\
    \textbf{\fontsize{22}{48}\selectfont \hsr@category}
    \vspace{0.5cm} \\
    \begin{tabular}{rl}
    \textbf{Version:} & \hsr@version \\
    \end{tabular}
    \vspace{4cm} \\
    \begin{tabular}{rl p{1.5cm} rl}
    \textbf{Autor:} & \textbf{\@author}         && Dozent:       & \hsr@instructor \\
    E-Mail:         & \texttt{\hsr@authoremail} && Themengebiet: & \hsr@topic \\
    Matrikelnummer: & \hsr@matriculationnr      && Studiengang:  & \hsr@course \\
    Adresse:        & \hsr@address              &&                 & \\
    \end{tabular}{}
    \vspace{2cm} \\
    \begin{tabular}{rl}
    \textbf{Erstellt am:}       & \textbf{\hsr@startdate} \\
    \textbf{Letze Änderung am:} & \textbf{\hsr@finishdate} \\
    \end{tabular}
    \end{center}
\end{titlepage}}

\renewenvironment{abstract}{%
    \section*{Abstract}%
    \newcommand\problem{\paragraph{Problem}}
    \newcommand\goal{\paragraph{Ziel}}
    \newcommand\method{\paragraph{Method / Vorgehen}}
```

```
    \newcommand\results{\paragraph{Wesentliche Ergebnisse}}
    \newcommand\suggestions{\paragraph{Empfehlungen}}
    \newcommand\keywords{\paragraph{Schl\"usselw\"orter}}
}{}

%% TODO: twoside / book support
%% Debug:
%% \RequirePackage{showframe}

\newgeometry{
    hmargin=\hsr@hmarginwidth,
    vmargin=\hsr@vmarginheight,
    marginparsep=0pt,
    marginpar=0pt,
}

%% Configure headers
\fancypagestyle{hsrbericht}{%
    \fancyhf{} % clear
    \fancyhead[C]{\@title}
    \fancyhead[R]{\hsr@clientlogo}
    \fancyhead[L]{\hsr@logo}
    \fancyfoot[R]{\thepage / \pageref{TotPages} }
    \fancyfoot[L]{Dockument: \jobname \\ Druck: \finishdate}
    \renewcommand{\headrulewidth}{.5pt}
    \renewcommand{\footrulewidth}{.5pt}
}

\fancypagestyle{plain}{%
    \fancyhf{} % clear
    \fancyhead[C]{\@title}
    \fancyhead[R]{\hsr@clientlogo}
    \fancyhead[L]{\hsr@logo}
    \fancyfoot[R]{\thepage / \pageref{TotPages} }
    \fancyfoot[L]{Dockument: \jobname \\ Druck: \finishdate}
    \renewcommand{\headrulewidth}{.5pt}
    \renewcommand{\footrulewidth}{.5pt}
}

\pagestyle{hsrbericht}

⟨/hsrbericht⟩
```